

Gao Haiyang

May 22, 2004



Personal Homepage: <https://haiyang-gao.github.io/>

Tel: (86) 185 1060 4219 Email: highsea_econ@foxmail.com

Educational Background

Imperial College London

09/2026-09/2027

Degree: Master of Science (in progress)

Major: Artificial Intelligence

Wuhan University Honor Graduate of Hongyi Honor College

09/2022-06/2026

Degree: Bachelor of Economics & Bachelor of Mathematics

Honours: Honor Graduate of Hongyi Honor College

Major: Finance

Minor: Mathematics and Applied Mathematics

Mark: 3.69/4.0 (Major)

Core Modules: (Numbers in the bracket is course grade out of 100)

Real Analysis (95), Numerical Analysis (95), Stochastic Process (91), Methods of Data Analysis (91), Machine Learning (90), Monetary Banking (96), Corporate Finance (95), Options, Futures and other Derivatives (93), Theory of Markets and Equilibrium (90)

Yale Summer Session

06/2025-08/2025

Courses: Game Theory (course grade: A), YData: Introduction to Data Science (course grade: A)

Published Papers

2024 Green Finance Annual Report, 2025, Second author, Industry Publication

Academic Projects

Sparseformer: Improving Long-Range Dependency Modeling for Time-Series Forecasting 07/2025-03/2026

Position: Researcher, supervised by Dr. Shivam Sharma, Yale University

- Led the design and manuscript drafting of a Sparseformer model for time-series forecasting, addressing vanilla Transformer's $O(L^2)$ self-attention cost, memory bottlenecks, slow long-horizon inference, and limited effective receptive fields through Transformable Sparse Attention with offset-network sampling and STL seasonal-trend decomposition features.
- Built the PyTorch training and evaluation pipeline on ETTh1, WTH, and ECL, benchmarked against Informer, Reformer, and DeepAR with MSE/MAE, and achieved consistently higher forecasting accuracy.
- Manuscript available [Here](#)

X-KANformer: A Cross-Attentive Kolmogorov–Arnold Transformer for Portfolio Allocation 03/2026–08/2026

Position: Researcher, supervised by Dr. Shivam Sharma, Yale University

- Led the model design of X-KANformer, an encoder-only financial time-series architecture that replaces the conventional Transformer feed-forward network with Kolmogorov–Arnold Networks and integrates endogenous stock-return patches with exogenous Fama–French factors and news sentiment through self-attention and cross-attention.
- Built the PyTorch training and evaluation pipeline for direct portfolio-weight learning under a negative Sharpe-ratio objective, incorporating selective learning to reduce the influence of anomalous financial observations; achieved an out-of-sample Sharpe ratio of 1.375 and ranked among the top five models on the JKP Common Task Framework leaderboard.
- manuscript available [Here](#).

Economic Effects of LA 2028 Olympics

08/2025-03/2026

Position: Research Assistant, supervised by Dr. Lorena Martin, Trojans Sports Research Lab, USC

- Participating in a research project on analysing the effect of LA 2028 Olympics to local economy.

Corporate Carbon Finance Product Innovation and Pricing

03/2025-07/2025

Position: Research Assistant

Sponsor: International Institute of Green Finance (IIGF), CUF, and State Grid Research Institute

- Contributed quantitative finance analysis to product-pricing discussions through weekly research meetings.

Platform Economy Research

09/2024-07/2025

Position: Research Assistant

Sponsor: Professor Shen Bo's Research Group, Wuhan University

- Collected market materials and analysed platform business models, competition, user behaviour, effects of advertising and its implications for consumer and social welfare.

Teaching Assistant for Game Theory Module in Wuhan University

02/2025-07/2025

Position: Teaching Assistant

- Assisted instructor in Game Theory course, collected and graded assignments monthly, selected as an outstanding TA, earning high recognition from both the instructor and students.
- Provided in-depth explanations during exercise classes, including addressing students' questions on Cournot equilibrium and other core models to strengthen their understanding of theoretical concepts.
- Organised Q & A sessions before quizzes and exams; supervised and graded mid-term and final exams, as well as collecting and grading monthly assignments.

Industry Experience (Internships)**China Telecom BestPay, Paid, Full-Time**

07/2026-09/2026

Position: Intern, Technology and Big Data Platform Department

- Engineered reusable LLM Skills that automatically ingest, structure, and update financial-violation records in a searchable database.
- Developed an enterprise risk model that combines LLM-derived violation signals with transaction, behavioural, and financial data to automate company risk classification.

Industrial and Commercial Bank of China, Paid, Full-Time

08/2025-10/2025

Position: Intern, Corporate Finance Department & International Business Department

- Assessed corporate credit applications and trade-finance documentation to identify market, financial, and transaction risks, supporting loan approval and cross-border financing decisions.

China Securities Co., Ltd, Full-Time

07/2024-08/2024

Position: Intern, Bond Underwriting Department

- Automated weekly bond-report data processing with Python and Wind, improving production efficiency by 40% and delivering 5 issues for market and partnership analysis.

KPMG Huazhen LLP, Paid, Full-Time

01/2024-02/2024

Position: Intern, Audit Department

- Built Excel templates for CRRC's annual audit to automate data import and validation, improving team efficiency by 70% while supporting evidence reconciliation and risk review.

Languages

Mandarin Chinese (native), English (proficient)

IELTS: Overall: 7.5 (Listening: 7.5, Reading: 9.0, Writing: 6.5, Speaking: 7.5)

Programming Skills

Python, The R Programming Language, STATA, LaTeX, MATLAB, Excel

Project Experience**Self-developed WeChat Mini Program**

A badminton-focused mini program built independently for match recording, Elo ratings, ranking comparison, and performance trend tracking. Currently have more than 700 users.

**Honours & Awards**

Outstanding Student of Hongyi Honor College for academic year 2024-2025

2nd class scholarship for academic year 2024-2025

3rd class scholarship for academic year 2023-2024

3rd Prize in National ETF Simulation Investment Elite Challenge

Outstanding Graduation Thesis for academic year 2025-2026

Won team championship 2 times as a key player, 2nd Place in Men's Doubles and 2nd Place in Mixed Doubles, Hongyi Honor College Badminton Tournament

3rd Place in university-level football tournament, MVP; 1st Place in college-level football tournament, MVP and Golden Glove Award